Available online at www.sciencedirect.com

ScienceDirect

IFAC PapersOnLine 54-1 (2021) 702–708

An AI adoption model for SMEs: a conceptual framework

Andrea Bettoni*. Davide Matteri*, Elias Montini*, Bartłomiej Gładysz, Emanuele Carpanzano***

**University of Applied Science and Arts of Southern Switzerland, Manno (e-mail: andrea.bettoni@supsi.ch, davide.matteri@supsi.ch, elias.montini@supsi.ch, emanuele.carpanzano@supsi.ch)*

***Warsaw University of Technology, Warsaw (e-mail: bartlomiej.gladysz@pw.edu.pl)*

Abstract: today literature proposes several models to assess the level of digitisation of a company. However, digitisation includes innumerable elements and aspects that require either models that are too complex to be easily applied by Small and Medium Enterprises (SMEs) or too high-level to provide significant hints for improvement. This paper proposes a model for measuring Artificial Intelligence (AI) readiness and promoting its adoption in SMEs. By focusing the approach, a model is obtained that is easy to apply, also for SMEs, and sufficiently detailed to provide relevant information and guidelines. The model has been already applied in a sample of 39 companies. It could serve for organizations to assess themselves and for authorities, industrial organisations and academia to diagnose selected populations (e.g. clusters, sectors, economies).

Keywords: Artificial intelligence, digitisation, manufacturing, technology readiness, small and medium enterprises, survey.

1. INTRODUCTION

The huge transformation brought by the fourth industrial revolution into the manufacturing world has forced any company, no matter the size, sector or location, to take on the digitisation journey. According to McKinsey Global Survey, the adoption of Artificial Intelligence (AI) is rapidly taking hold across global business (McKinsey, 2018). In particular, respondents from different functions highlight that the greatest value in using AI is seen in manufacturing than in other sectors. In fact, several AI methods have been applied in the manufacturing context including Machine Learning, Deep Learning, Convolutional Neural Networks and many others to solve tasks of different nature such as predictive maintenance, process optimisation, computer vision, etc. (Fahle, Prinz, & Kuhlenkötter, 2020). However, still manufacturing companies, in particular SMEs, are struggling in understanding the real impact that AI can have on their business and the requirements to introduce this technology (Gartner, 2018).

It is up to managers to understand the feasibility and benefits of adopting AI solutions. However, only 18% of them state that their companies have a clear strategy in place to collect data and use AI. Lack of strategy is, together with the lack of competences, lack of a governance model for implementation and lack of constrained end-to-end AI solutions, the most critical barrier towards AI adoption (McKinsey, 2018). These barriers are more challenging for SMEs (Jabłońska & Pólkowski, 2017) because:

- AI solutions require investments and resources;
- AI-based implementations need to work on well-prepared data from credible sources.

SMEs usually lack both of these elements. However, due to the high relevance of SMEs in the European economy, a two-speed transformation that leaves them behind is not affordable. Currently, SMEs are particularly slow in integrating digital technologies: only one out of five SMEs in the EU is highly digitised, although SMEs represent over 99% of all businesses in Europe, they account for two out of every three jobs and for 56.4 % of value-added within the EU non-financial business economy (European Commission, 2020).

Despite all these challenges, the benefits and the competitive advantage that this technology can bring are undeniable (Mills, 2019; Otar, 2019). It is necessary to consider that SMEs have historically been less enthusiastic about new technologies (How, 2018). If a SME rushes with the adoption of AI and it eventually proves to fall short of its expectations, it can hardly manage time, resources and know-how invested.

On one side, in the past five years, SMEs have created around 85% of new jobs and provided two-thirds of the total private sector employment in the EU, representing a key to ensure economic growth, innovation, job creation, and social integration in the EU. On the other side, they represent a small part of European investments for the propagation of innovative technologies. They rely much more on external innovation, integrating new technology when it is at its ripe, reducing the investment risk at the expense of losing the first-mover advantage.

However, technologies supporting AI are in continuous evolution, becoming achievable for many SMEs, thus facilitating the adoption of solutions previously reserved only for large organizations due to the huge financial disbursements they required. Therefore, SMEs' CEOs and managers need models and tools to support the adoption of AI solutions. These models should provide guidelines and assessment methods to

Peer review under responsibility of International Federation of Automatic Control.
10.1016/j.ifacol.2021.08.082

guide the decision-making process also in the absence of deep knowledge of the technology. Despite the several methods and framework to develop AI applications, few and very limited models exist that support managers in adopting AI at organisational level, making this technology a competitive advantage. Moreover, these models require good competences in the field of digitalisation and AI, making their application complex for non-experts. On the contrary, the model proposed in this paper, and its connected questionnaire, allow companies to easily self-assess their maturity level and provide guidelines to either start or feed their AI journey.

The model is composed of five pillars and supports companies in evaluating their AI readiness through a specific questionnaire and assessment method. Moreover, it provides a set of guidelines to support the introduction of AI solutions in production systems. The method targets CEO's and managers, and not AI and digitalisation experts, who can be hardly found in SMEs, proposing specific questions and guidelines covering the whole organisational aspects, without focusing on deep technical elements.

In section 2, the method applied to develop the model and its main elements are introduced, including a review of the existent models focusing on the evaluation of AI readiness and maturity. In section 3, the model is presented, together with its pillars and the guidelines to support SMEs in AI adoption. Section 4 presents the results of a survey built on the questionnaire to assess AI readiness in a target sample of European companies. The research has a piloting nature. Its goal is to formulate further hypothesis and to test the questionnaire. For this reason, the selection of the sample is not representative but aimed at maximising the number of respondents. Finally, the last section introduces the future steps of this research.

2. THE METHOD APPLIED TO DEVELOP THE AI ADOPTION MODEL FOR SMEs

The work carried out in this research follows the methodology presented in Fig. 1. The proposed model has been developed starting from a review of the state of the art and direct interviews involving SMEs' managers. These activities allowed to identify the existent gaps in models and tools that should support companies in evaluating their AI readiness and level of adoption. From these gaps, the adoption model with its assessment method and guiding questions have been designed and developed.

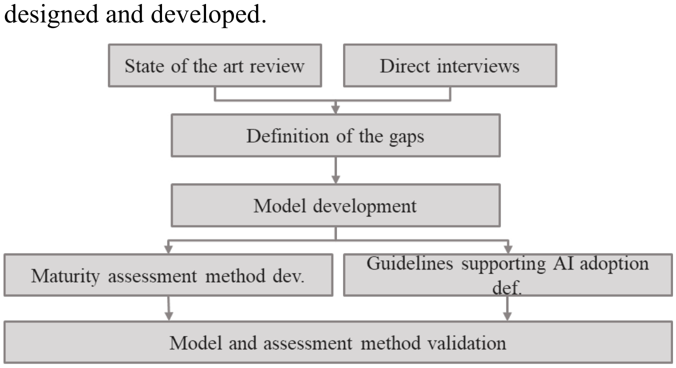

Fig. 1 Research method

2.1 Direct interviews

Companies must address significant challenges and develop internal capabilities if they want to implement and adopt AI-based solutions. In order to identify the most prominent challenges that a company faces implementing these solutions, how these challenges are addressed, and how employees react to AI adoption, a state of the art review has been carried out and complemented with direct interviews. Two managers from two SMEs have been directly interviewed. One company produces battery packs and it is located in Poland. The other designs and develops machines for inspection and sorting and it is located in Italy. Even though some of the questions proposed in these interviews have been reused in the questionnaire presented in section 4, the main goal of having a face-to-face talk with industrial experts was to validate the exhaustiveness of the designed analysis.

According to the interviewees' experience, it is mainly the poor level of digitisation in SMEs that does not allow to apply AI solutions. Moreover, according to the results of the face-to-face interviews and to the review of the state of the art (Raj, Dwivedi, Sharma, L., & Rajak, 2020), (Gartner, 2019), (Forbes, 2019), five key barriers prevent companies in the adoption of AI:

- **Lack of data**: companies do not have enough data to feed AI. Many are still collecting data manually or not collecting them at all. Since AI needs huge amounts of data, reliable and continuous data collection and storage is fundamental to find patterns and train AI systems. Without data, it is very difficult for algorithms to discover the desired insights leading the company to resolve problems. Therefore, structuring and automating data collection has to be a priority for companies willing to adopt AI.
- **Lack of AI lifecycle assessment methods**: companies have the perception that the cost of AI solutions is still too high. This could be true in many cases, but it should be compared with the achievable benefits. However, there is a lack of AI-specific methods and tools to estimate the cost/advantage ratio and the Return On Investment. Moreover, companies struggle in defining a clear path to AI adoption making the definition of investment risks complex.
- **Lack of customised solutions**: companies struggle in finding personalized AI solutions that can solve their problems at an affordable price. AI systems on the existing marketplaces, or those directly offered by software houses may miss features that companies need or, to the contrary, offer too many features that increase the price and remain unused.
- **Under-skilled employees**: employees are not ready to the AI transformation. They need more training for producing value for the company out of AI usage. Digital skills are important and having a strategy for measuring the current gap and for providing the needed upskilling is an unescapable requirement.
- **Complexity in using solutions**: AI tools are still too complicated. Companies need simple systems that can be deployed and used quickly by employees

avoiding complex setups, commands or interactions. An easy-to-use system can avoid non-value-added activities by adopting simply operations to obtain a clear and expected result.

2.2 AI maturity models – state of the art

Investigating recent studies in the field of AI for industry, the results on AI maturity models are very limited. Many models have been developed in the context of Industry 4.0, as presented by Channias and Hess (2016) and by Canetta et al. (2018), but few are specific on AI or even consider it.

Among these, the work of Zhang et al. (2019) proposes a reference framework of AI for industry. This framework, together with the classification of AI's functions, technologies, domains and objects, includes a classification of value chain stages and applications for AI adoption in industry. However, this classification structure presents limits especially in guiding companies in the identification of possible applications in their industrial context. These are described at a high level which does not allow to really understand which could be the role that AI performs.

Gartner (2019) has developed an AI maturity model that segments companies into five maturity levels (from least to most mature) regarding an organization's use of AI:

1. Awareness: early AI interest with risk of overhyping;
2. Active: AI experimentation mostly in data science context;
3. Operational: AI in production creating value by process optimization or product/service innovations;
4. Systemic: AI is pervasively used for digital process and chain transformation, and disruptive new digital business models;
5. Transformational: AI is part of business DNA.

Based on interviews with various industry experts, Altimeter (2018) has created its own maturity model, identifying four maturity stages (from least to most mature):

1. Exploring: company is exploring AI, engaging with experts, considering use cases, but has not yet committed significant time or resources;
2. Experimenting: company is actively experimenting with AI for a range of use cases, using either internal (employee) or external (service) resources;
3. Formalizing: AI is becoming a formal part of corporate strategy, and data is now a core competency across the organization. Implementations are moving beyond optimization;
6. Integrating: AI is part of the fabric of the company, embedded into processes, products and services across the organization, and is delivering value across the business.

The Altimeter's model has more specific elements compared with Gartner's one. Specifically, for each stage, it provides the description of the five relevant organization areas to be evaluated (Strategy, Data science, Ethics and governance, Product & Service development, Organization & Culture).

Similar to the previous one is the Element's AI maturity model (Ramakrishnan, 2019), structured in 5 stages. The first three are Exploring, Experimenting and Formalizing, as for the Altemeter's model. The last stages, that in the Altemeter's model correspond to 'Integrating', are:

- Optimizing: scaling AI solution deployments \efficiently as the number of deployed AI models increases;
- Transforming: transforming the organization itself through the use of AI. The organization uses AI in how it operates across many critical areas of the business.

Also in this model, five pillars have been defined (Strategy, Data, Technology, People and Governance) to evaluate AI maturity in a systematic approach.

Regarding maturity models that are not specific to AI, it is worth mentioning the most recognised one in the context of digitisation: the "Acatech Industrie 4.0 Maturity Index" (Schuh, Anderl, Gausemeier, Hompel, & Wahlster, 2017). This model provides the means for establishing companies' current Industry 4.0 maturity, supporting also the identification of concrete measures to achieve a higher maturity level. Although it does not deal directly with AI, it is the most consolidated model on digitisation, which is the prerequisite for the introduction of AI. In fact, the Acatech model places its emphasis on digitisation and on the themes necessary for a generic company to tackle the path towards complete automation of its processes in the best possible way without referring to AI as the main promoter.

Finally, it is worth mentioning that all these maturity models are generically conceived for any company regardless of its size. Even though the steps they focus on are applicable also for SMEs, they operate more at a strategic level without offering practical support for AI adoption.

3. THE AI MATURITY AND ADOPTION MODEL

This section describes the model proposed in this research, specifying the organization's areas that influence and are influenced, directly or not, by the adoption of AI. The model aims to go beyond existing ones, providing not only the pillars for AI adoption in industrial companies but also operational tools to support them in the transformation. For this reason, the model substantially follows the approach of its predecessors yet providing additional features:

- The maturity level is not based on a single dimension but it quantitatively evaluates different areas (pillars) relevant for AI in order to identify where a company has to improve to be ready for AI adoption and get the maximum out of it;
- A specific assessment, based on a questionnaire, returns the level of maturity for each pillar, allowing a company to identify the most underdeveloped pillars;
- The model is oriented to SMEs but usable by all, also big companies, thanks to its general structure, easy

applicability in different context and sectors without requiring specific competences on AI;

- It includes a set of guidelines aiming at supporting manufacturing SMEs' CEOs and managers to understand what they have to consider for adopting AI solutions and exploiting their complete potential.

The model, represented in Fig. 2, focuses not only on the operational aspects but also at a higher level, on how organization and culture inside a company allow it to fully exploit the potential of AI. The model is structured in five pillars, each representing a company area that needs to be developed for the adoption of AI. These pillars have been defined by analysing the core areas of a company, identifying those that influence or that are influenced by AI adoption. The pillars are:

- Digital and Smart Factory;
- Data Strategy;
- Human Resources;
- Organizational Structure;
- Organization's Culture.

For each pillar, a brief discussion is presented along with the best practices that every company needs to adopt and the elements to be considered for AI adoption.

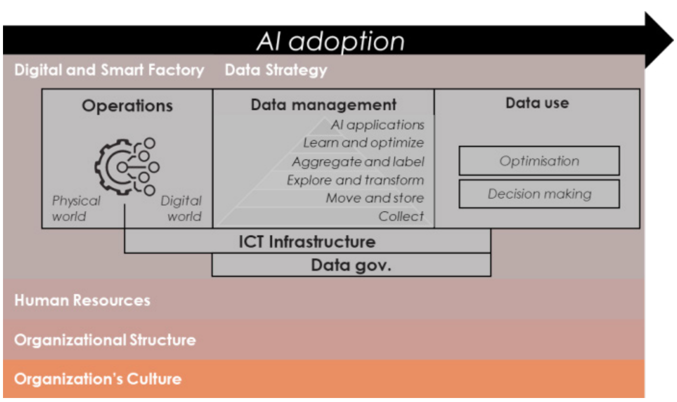

Fig. 2 AI maturity and adoption model

3.1 Digital and Smart Factory

Internet of things (IoT) and Industry 4.0 bring in the manufacturing context the concept of digital factory, where physical objects in the shop floor, people and systems are digitalized to build a digital representation of the factory (Shariat Zadeh, Lundholm, Lindberg, & Franzén Sivard, 2016). The Smart Factory is a step beyond where successful implementation of Industry 4.0 enables distributed and highly automated production. According to Yoon et al. (2012), a Smart Factory is a factory system in which autonomous and sustainable production takes place by gathering, exchanging and using information transparently anywhere anytime with networked interaction between man, machine, materials and systems, based on ubiquitous technology and manufacturing technology. The Digital Factory allows, on the one hand, to collect data in order to have a unique and real-time representation while the Smart Factory, on the other hand, has to implement decisions and to perform optimizations that are defined by decision-makers, including AI.

Digital representation can be realized by collecting data from information systems (e.g. PLC, CIM, ERP, MES, etc.) or directly through sensors. The latter are mostly used to monitor processes and any short-term adjustments applied to them. In today's industries, it is increasingly important to have sensors and resources that are capable of digitising reality so that a computer can analyse these data and present workable solutions to increase performance (Alcácer & Cruz-Machado, 2019). Thanks to digitisation of the real world, information can be displayed, analysed, stored and used to make better decisions that generate benefits for operations and business (Rasheed, San, & Kvamsdal, 2020). It is properly when AI is used to support or even it is in charge of decision making that it brings its maximum potential and value to companies. However, it is necessary to consider that to implement autonomous decisions is an ambitious objective that cannot be always realized at factory level, though at lower level, many successful examples exist (Bettoni, et al., 2020; Mescoli, 2019).

3.2 Data strategy

Data is essentially the plain facts and statistics collected during operations. AI solutions take data from the factory as input and are able to extract useful information, optimize operations and suggest or even make decisions.

Without data, AI cannot be implemented. Data can be used to measure and record a wide range of business activities and, in particular, operations. While data themselves may not be very informative, they are the basis for all AI solutions. The increasing use of sensors, smart machines and connected objects across the shop floor is making the volume of data generated grow every second. However, it is necessary to consider that data has to be easily accessible, sometimes also in quasi-real time, filtered, stored and labelled (Rogati, 2016). Moreover, data management has to be supported by procedures that raise the need for companies to have proper data governance and proper infrastructure.

Finally, to transform data in valuable assets, it is fundamental the data governance. This includes availability, accountability, usability, consistency, data integrity and data security. It includes establishing processes, procedures and policies to ensure effective data management (Janssena, Brousa, Estevez, Barbosa, & Janowski, 2020). Moreover, the ICT infrastructure, which is also strictly related with the Digital and Smart Factory pillar, is fundamental in order to manage and use data.

3.3 Human Resources

As AI solutions spread through company processes, the workforce must possess certain competencies in order to make the most out of the collected information. According to Matsa and Gullmajji (2019), AI integration into workers' practices allows to make better decisions, getting the full potential from both AI and workers. AI enables workers to focus on what they do best or just support them in doing it more efficiently, delegating low value-added tasks to AI and be more productive in their main tasks.

Companies must also help their employees to acquire integrated, interdisciplinary IT skills that provide them with a basic understanding of the applications and processes used in different parts of the business, thereby upgrading long-established occupational profiles.

Moreover, the confidence between workforce and AI outputs is very important. Explainability is a prerequisite for building trust and adoption of AI systems (Gade, Geyik, Kenthapadi, Mithal, & Taly, 2019) (Gade, Geyik, Kenthapadi, Mithal, & Taly, 2020) because it allows employees to understand how decisions are made and recognize their concrete benefits.

3.4 Organizational structure

The adoption of AI inside a company requires, as most of the ICT projects, the right organizational structure. The transition to AI-based organizations involves a left-to-right rethinking of how organizations run their operations, engage their customers, empower their employees, and even define their products.

The implementation of a new technology can lead many factors to make organizational change necessary. With AI tools and their prescriptive capability, the organizational change should be mainly focused on continuous improvement, preferring adaptive changes to transformational ones. Typically, these changes are minor modifications and adjustments that managers fine-tune and implement to execute upon business strategies. Moreover, AI needs investments and resources. Therefore, budget has to be correctly managed in order to start pilot initiatives, in case no solutions are already present, or strategic projects, if AI is a key factor for company future business success.

3.5 Organization culture

A company's agility is highly dependent on the behaviour of its employees. Experience with lean management in the 1990s and 2000s showed that the key to its successful implementation throughout a business is to change the company's culture – in other words, to change the mentality of its employees (Liker, 2004). The same applies to the digital transformation and AI adoption. Companies will be unable to achieve the benefits if they simply introduce technologies without also addressing their corporate culture. AI demands a culture shift because it alters the relationship between machines and humans, changing machines from passive receivers of commands into informed, sentient collaborators. In doing so, it will also transform the skills that organizations seek to find and foster in human workers.

Productivity benefits of digital technologies like AI depend on the adopting firms' ability to reorganize their processes around the new technology. This sort of transformation is more likely in cultures that promote experimentation, learning agility and a growth mind-set (Lu & Ramamurthy, 2011). A company's culture that promotes the experimentation, trying to innovate, is a better approach for the introduction of AI solutions.

4. AN ASSESSMENT METHOD TO QUANTIFY AI MATURITY LEVEL

The maturity assessment method evaluates each of the pillar composing the AI adoption model presented in the previous section and aims to support SMEs in fully developing those areas. The assessment relies on a questionnaire structured in three sections:

- **A section to classify the company**: this section allows to collect company characteristics like size, class of the company, the business sectors.
- **A section for the company's attitude to AI**: in this section, it is investigated the company's attitude towards AI, the benefits that the company has gained from its adoption (or that it estimates or believe it will achieve) and problems that prevent it from making the most out of its potential.
- **A section dedicated to the analysis of each pillar of the model**: this section investigates on the proposed pillars to understand the current situation of the company in relation to them by returning a score for each one.

A set of predefined questions compose each section. Each question refers to a specific pillar (p) and has a specific weight (W_i). W_i have been defined by the authors, based on their expertise in the field, assessing questions and answers relevance on AI readiness and adoption. The answers to each question contribute with different scores ($S_{i,j}$) based on the best practices for the topic analysed in the question. For each pillar p, there are q_p questions and a_p answers. The weighted scores contribute to calculate the maturity score for each pillar (S_p). The formula to calculate the score for each pillar can be sum-up as follow:

$$S_p = \frac{\sum_{i=1}^{q_p} W_i \cdot \sum_{j=1}^{a_p} Y_j \cdot S_{i,j}}{\sum_{i=1}^{q_p} W_i \cdot \max_j S_{i,j}} \cdot 100 \qquad \forall p$$

where Y_j is 1 if the answer has been selected by the interviewed, or 0 if it is not.

This formula allows to calculate a quantitative score that supports a company in understanding which are the weak areas currently preventing AI adoption or its full exploitation. This assessment method has been applied through a survey involving 39 companies from different European countries, 30 SMEs (companies which employ fewer than 250 persons and which have an annual turnover not exceeding EUR 50 million) and 9 large enterprises as a control group to verify if significant differences appear in their approach to AI.

From the analysis of the results, the radar chart represented in Fig. 3 has been drawn. It shows, for each pillar, the average maturity level (0-100) of SMEs and large enterprises groups answering the survey. Considering surveyed sample and pilot nature of this survey, some hypotheses were formulated for further testing:

- Regardless of the size, companies are unprepared to adopt AI (low levels in every dimension).
- The maturity levels in SMEs and large enterprises related with organization culture and human resources are similar, and both have a significant gap to be covered.
- Digital & Smart Factory and organizational structure present a relevant difference between the two types of surveyed.
- Maturity level of Digital & Smart Factory is significantly higher for large enterprises, however, both samples have significant room for improvement.
- SMEs are more prepared on data strategy because they started by wondering what AI and digitisation can bring to them, while large enterprises many times force the adoption of technologies to follow trends. However, SMEs have not a ready organizational structure and for this reason, they are weaker on the Digital and Smart Factory pillar.

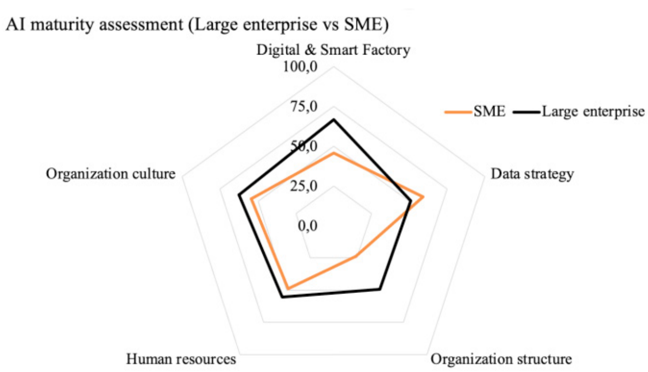

Fig. 3 AI maturity assessment: SME vs Large Enterprise

The use of the model allowed to elaborate specific questions for each pillar to formulate an assessment that considers all the fundamental aspects towards the adoption of AI. Thanks to this first application, it has been possible to refine the questions and the evaluation system in order to obtain a more precise and complete assessment. This constitutes the base for further research, including refined assessment, greater sample (to identify phenomena and correlations) and qualitative in-depth interviews (to understand phenomena and correlations).

5. CONCLUSION

This research provides a conceptual framework to support AI adoption in SMEs. The analysis, carried out on existing models from literature and through interviews to manufacturing SME's managers, allowed to identify the current gaps that prevent SMEs from developing their AI systems using the current models as guidelines. Specifically, models are missing for SME-accessible self-assessment and provision of recommendations both at strategic and operational level.

Based on this analysis, the proposed framework and its main pillars were defined. Moreover, the presented work designed an assessment method to measure each pillar's development level that has been concretized in a survey for CEOs and managers in manufacturing companies. The results of a first sample represent the starting point for:

- A wider investigation on AI readiness and adoption in European SMEs to identify correlations among the characteristics of the company and the achieved maturity levels.
- The identification of maturity archetypes for designing generic and reusable patterns for AI adoption based on the starting point and on the company's contingent factors.
- The inclusion of further elements like ethics to improve the exhaustiveness of the model.
- A further breakdown of pillars into the desired features that the company should show in each area, so allowing to provide a more detailed assessment and to build specific recommendations for improvement.
- A more accurate scoring method achieved by means of the review of the weights (W_i) through the interview of different AI experts applying AHP method.
- An adaptive questionnaire allowing to select the questions to be proposed to the respondent on the basis of the answers already provided with the aim to achieve more sound scores.

All these actions combined are meant to extend the reliability of the insights generated by the survey responders. Currently, the main limitations regard the relatively small sample that participated in the survey and the consequent low coverage of the industrial sectors and industrial regions.

A more refined and robust survey could become a valuable tool in the hands of manufacturing companies' managerial figures for self-assessment of their positioning towards AI and, at the same time, industrial associations and clusters could use it as a service to their members.

AKNOWLEDGMENT

The work presented in this paper was developed within the project "KITT4SME - platform-enabled KITs of arTificial intelligence FOR an easy uptake by SMEs" and received funding from the European Union's Horizon 2020 research and innovation programme under grant agreements No 952119.

REFERENCES

Alcácer, V., & Cruz-Machado, V. (2019). Scanning the industry 4.0: A literature review on technologies for manufacturing systems. *Engineering Science and Technology an International Journal, 22*(3), 899-919.

Altimeter. (2018). *The AI Maturity Playbook: Five Pillars of Enterprise Success.* Prophet.

Bettoni, A., Montini, E., Righi, M., Villani, V., Tsvetanov, R., Borgia, S., . . . Carpanzano, E. (2020). Mutualistic and Adaptive Human-Machine Collaboration Based on Machine Learning in an Injection Moulding Manufacturing Line. *Procedia CIRP*.

Canetta, L., Barni, A., & Montini, E. (2018). Development of a Digitalization Maturity Model for the manufacturing sector. In T. a. IEEE International Conference on Engineering (Ed.)., (pp. 1-7).

Chanias, S., & Hess, T. (2016). How digital are we? Maturity models for the assessment of a company´s status in the digital transformation. *Munich Sch. Manag., 2*.

European Commission. (2020). *Digital Innovation Hubs (DIHs) in Europe.* Retrieved from https://ec.europa.eu/eurostat/documents/3217494/11432756/KS-EI-20-001-EN-N.pdf/6b9097d9-ea05-a973-d931-c08334db979b?t=1602776043000

Fahle, S., Prinz, C., & Kuhlenkötter, B. (2020). Systematic review on machine learning (ML) methods for manufacturing processes–Identifying artificial intelligence (AI) methods for field application. *Procedia CIRP*, 413-418.

Forbes. (2019). *The 4 Biggest Barriers To AI Adoption Every Business Needs To Tackle.*

Gade, K., Geyik, S. C., Kenthapadi, K., Mithal, V., & Taly, A. (2020). Explainable AI in industry: practical challenges and lessons learned: implications tutorial. Conference on Fairness, Accountability, and Transparency.

Gartner. (2018). *Applying Artificial Intelligence to Drive Business Transformation: A GartnerTrend Insight Report.*

Gartner. (2019). *3 Barriers to AI Adoption.*

Gartner. (2019). *The CIO's Guide to Artificial Intelligence.*

How, J. N. (2018, August). SMEs Can Catch The AI Wave. Harvard Business Review. *Harvard Business Review*.

Jabłońska, M. R., & Pólkowski, Z. (2017). Artificial Intelligence-based processes in SMEs. *Studies & Proceedings of Polish Association for Knowledge Management, 86*.

Janssena, M., Brousa, P., Estevez, E., Barbosa, L. S., & Janowski, T. (2020). Data governance: Organizing data for trustworthy Artificial Intelligence. *Government Information Quarterly, 37*(3).

Liker, J. (2004). *The toyota way.* Esensi.

Lu, Y., & Ramamurthy, K. (. (2011). Understanding the Link Between Information Technology Capability and Organizational Agility: An Empirical Examination . *MIS Quarterly , 35*(4), 931-954.

Matsa, P., & Gullamajji, K. (2019). To Study Impact of Artificial Intelligence on Human Resource Management. *International Research Journal of Engineering and Technology(IRJET), 06*(08).

McKinsey. (2018). *AI adoption advances, but foundational barriers remain.*

McKinsey. (2018). *McKinsey Global Survey.*

Mescoli, F. (2019). *Artificial intelligence for future agile manufacturing.* Retrieved from https://www.kit.edu/kit/english/pi_2019_047_artificial-intelligence-for-future-agile-manufacturing.php

Mills, K. (2019). How AI Could Help Small Businesses. *Harvard Business Review*.

Otar, C. (2019, April). Four Ways Artificial Intelligence Can Help Your Small Business. *Forbes*.

Raj, A., Dwivedi, G., Sharma, A., L., d. S., & Rajak, S. (2020). Barriers to the adoption of industry 4.0 technologies in the manufacturing sector: An inter-country comparative perspective. *International Journal of Production Economics, 224*.

Ramakrishnan, K. (2019). *AI from exploring to transforming: Introducing the AI Maturity Framework.* Element AI.

Rasheed, A., San, 2. O., & Kvamsdal, T. (2020). Digital Twin: Values, Challenges and EnablersFrom a Modeling Perspective. *IEEE Access, 8*.

Rogati, M. (2016). *Data Science Hierarchy of Needs.*

Schuh, G., Anderl, R., Gausemeier, J., Hompel, M. t., & Wahlster, W. (2017). *Industrie 4.0 Maturity Index.* Acatech. Retrieved from https://www.acatech.de/wp-content/uploads/2018/03/acatech_STUDIE_Maturity_Index_eng_WEB.pdf

Shariat Zadeh, N., Lundholm, T., Lindberg, L., & Franzén Sivard, G. (2016). Integration of digital factory with smart factory based on Internet of Things. *26th CIRP Design Conference, 50*.

Yoon, J.-S., Shin, S.-J., & Suh, S.-H. (2012). A conceptual framework for the ubiquitous factory. *International Journal of Production Research, 50*, 2174–2189.

Zhang, X., Ming, X., Liu, Z., Yin, D., Chen, Z., & Chang, Y. (2019). A reference framework and overall planning of industrial artificial intelligence (I-AI) for new application scenarios. *The International Journal of Advanced Manufacturing Technology, 101*, 2367–2389.